Acoustic Optimization of Fan Blade Geometry

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Grade 11

Period: G

INTRODUCTION

The modern world is constantly changing. Technology has completely transformed our lives over the past 200 years, but it also brings with it technical, social, and other problems. Until recently, fans were not particularly relevant to most people's lives. However, with the advancement of technology, the advent and widespread use of civil aviation, the emergence of comfort standards, and the widespread use of personal computers and other household appliances that require cooling, fans and turbines have quietly become one of the main sources of noise in homes and large cities for most people.

The term "noise pollution" is a relatively recent development, but advances in medicine have made it clear that it is a serious problem that can significantly impact people's lives and health. Noise pollution is recognized as a factor that adversely affects sleep, cognitive performance, and learning (Pellegatti et al., 2023). This problem poses a serious challenge to modern engineering, as noise pollution can seriously affect human cognitive function, reducing concentration, memory, and sleep quality (Basner et al., 2017). Studies have shown that aircraft noise exposure is associated with poorer reading comprehension and poorer recognition memory in children (Stansfeld et al., 2005). Furthermore, noise pollution increases blood pressure and heart risks, with research demonstrating that the impact of airplane noise at night on cardiovascular disease includes the rise in blood

pressure, the increase of stress hormones, and the impairment of endothelial function (Itzkowitz et al., 2023). Noise pollution also affects the environment by frightening animals and disrupting their communication. This is also one of the barriers to the widespread adoption of aeromobiles and similar technologies in modern life.

For many years, engineers have struggled to solve this problem, creating various designs and sound-absorbing systems that reduce noise levels. In the vicinity of airports, aircraft noise is considered one of the most detrimental environmental effects of aviation, causing community annoyance, disrupting sleep, adversely affecting academic performance of children, and potentially increasing the risk for cardiovascular disease (Basner et al., 2014). However, noise pollution remains a major concern in the design of aircraft and HVAC systems. Beyond aviation, HVAC systems are essential for maintaining optimal indoor environmental conditions, yet their operational noise presents a significant challenge to occupant well-being and performance, with noise characterized by its dominance in the lower frequency spectrum (Chen et al., 2025).

This research introduces a systematic approach to fan blade acoustic optimization by combining design principles with experimental validation. While previous studies have addressed fan noise through passive damping materials or motor control systems (Gee & Sommerfeldt, 2014; Lee et al., 2005), this project focuses specifically on geometric modifications based on general aerodynamic principles such as vortex reduction, flow smoothing, and pressure distribution control. Research has demonstrated that

leading-edge modifications can facilitate the break-up of vortices passively and effectively without negatively impacting aerodynamic performance (Chen et al., 2021). These design approaches, inspired by observations of natural structures known for silent operation, can help engineers design quieter, more aerodynamic fans. Redesigned blades can minimize turbulent airflow and tonal noise, making them more suitable for urban applications such as drones, ventilation systems, and electric vehicles, where low-noise operation is crucial for social and environmental well-being.

Despite the extensive research on fan noise reduction, there remains a gap in understanding how specific geometric modifications to blade design—particularly those involving edge treatments, center closure, and multi-ring optimization—affect acoustic performance while maintaining aerodynamic efficiency. Most existing solutions focus on external noise damping or motor control rather than optimizing the fundamental blade geometry itself. This project aims to address this gap by systematically testing geometric modifications and quantifying their impact on noise generation.

Research Question: Can changes in fan blade geometry based on basic principles of aerodynamics reduce noise pollution while maintaining aerodynamic performance?

Hypothesis: By optimizing the shape and geometry of fan blades, it is possible to significantly reduce noise pollution without sacrificing airflow efficiency.

Variables:

Independent variable: Blade shapes and geometric modifications

Dependent variables: Sound pressure levels (dB), aerodynamic performance (thrust)

Controls: Motor and mounting system, test setup geometry, RPM (rotations per minute), ambient conditions, microphone type and placement, measurement distance and procedure, data averaging method

Overview of Methods: Multiple fan blade geometries were designed with modifications including optimized fitting, closed center designs, blade shape optimization, and second ring optimization. Each prototype was 3D-printed and tested for sound pressure levels at three measurement angles (0°, 45°, and 90°) at a standardized distance of 4 centimeters from the fan blades. Sound measurements were compared against a baseline prototype and a standard commercial fan to evaluate the effectiveness of geometric modifications.

METHODS

Materials

- 3D printer capable of printing with PLA filament (Cobra 2)
- 3D modeling software (Fusion 360)
- 3D printer slicing software for print preparation
- PLA plastic filament
- A small electric motor (120V, 3000 RPM)
- A 120V power outlet
- A cardboard box exceeding 20 inches in each dimension
- Noise-absorbing foam with a thickness of 3 centimeters
- Sound level meter application installed on a smartphone

Procedure

Fan Design and Fabrication: Five different fan blade geometries were developed based on aerodynamic principles. The baseline prototype featured standard blade geometry. The fitting was optimized by using several durable screws that held the fan securely during rotation at high speeds. In one of the modifications, the central part was modified to close

potentially turbulent zones. The blade edges were sharpened after 3D printing. The second ring borders were also sharpened to reduce turbulence.

The prototypes were designed using Fusion 360 and fabricated using a Cobra 2 3D printer with PLA filament at 10% infill settings. Each fan consisted of 13 blades with a diameter of 154 millimeters.

Test Setup: A test rig was constructed to mount the 3D-printed fans on a standard kitchen fan electric motor. The motor was powered by a 120V energy source and operated at a constant speed of 3000 rotations per minute. The test rig was also placed into a noise-reducing enclosure constructed from cardboard and foam to reduce background noise interference.

Sound Measurements: Each newly 3D-printed fan was mounted on the motor to perform noise measurements. Sound pressure levels were measured using a smartphone with sound measuring software. The sound level meter was positioned in three different locations relative to the fan: directly below the prototype at the axis (90° position relative to the fan blade plane), at a 45-degree angle, and at a 0-degree angle (parallel to the fan blade plane). For the standard commercial fan, measurements were recorded in the same order but from the upper hemisphere, since the blades rotated in the opposite direction. All measurements were conducted at a distance of 4 centimeters from the fan blades.

Although this distance lies in the acoustic near field, it was sufficient for relative comparison between fan geometries, which was the primary goal of this study.

For each fan configuration, three measurement trials were conducted and averaged to ensure consistency. Ambient temperature was maintained at approximately 25°C with relative humidity of 50%. Background noise levels were measured and recorded at 40 dB before each test session.

Thrust Measurements: Thrust was measured using a laboratory scale. The testing setup was placed on the scale to measure the change in weight of the setup while the fan was operating compared to when it was not operating. The difference in weight readings provided a measurement of the thrust generated by the fan.

RESULTS

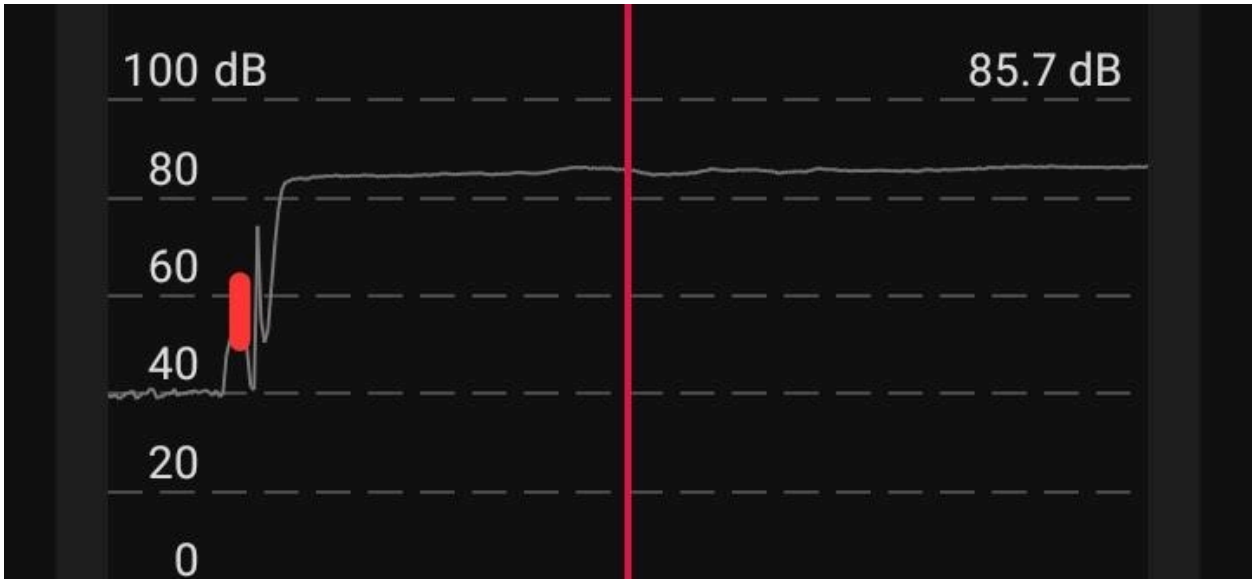
Sound Pressure Level Measurements

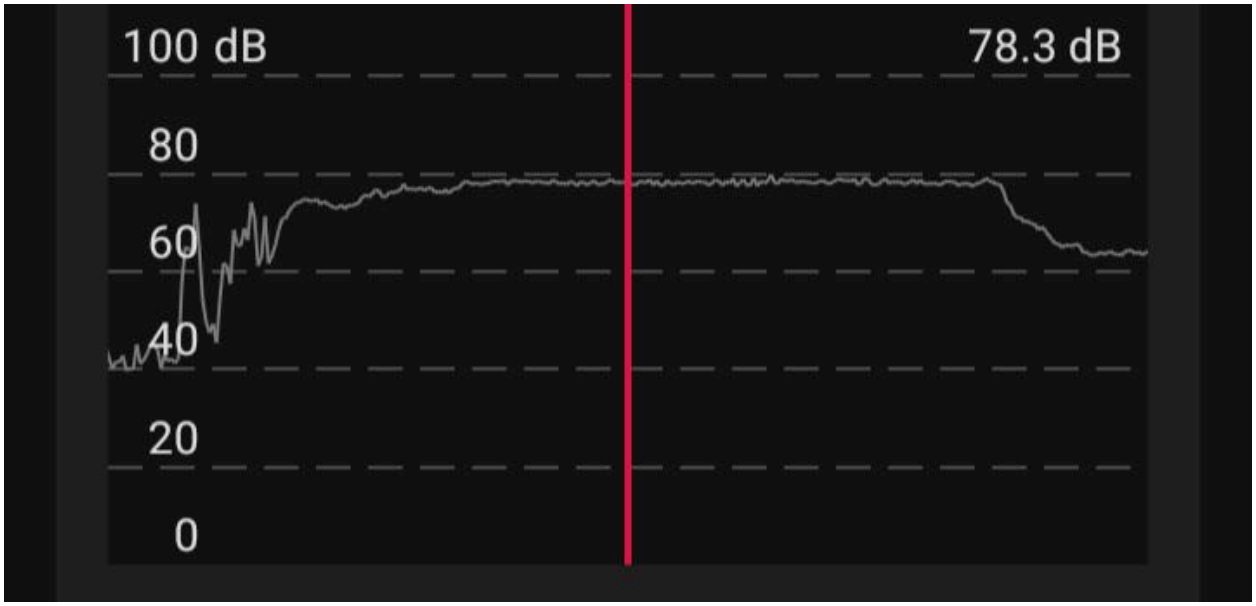
After refining the mounting system through four iterative phases, comprehensive tests were conducted on several modified versions of the original fan geometry. The table below represents the sound pressure levels measured at three different angles for each fan configuration.

Table 1: Sound Pressure Levels (dB) at Different Measurement Angles

Blade Design Configuration	0° Axial	45° Angle	90° Angle	Average
Base Prototype	80.0	79.0	77.5	78.8
Optimized Fitting	79.5	79.5	76.5	78.5
Closed Center	79.5	79.5	76.5	78.5
Blade Shape Optimization	78.5	78.5	76.0	77.7
Second Ring+Blade Shape Optimization+optimized fitting	78.0	78.5	76.0	77.5
Standard Commercial Fan	80.0	86.0	75.5	80.5

Note: All measurements in decibels (dB) taken at 4 cm distance from fan blades at 3000 RPM





Commercial and modified prototype sound diagrams

Qualitative and Quantitative Analysis

The baseline prototype produced sound pressure levels averaging 78.8 dB across all three measurement positions, with values of 80.0 dB at the axial position, 79.0 dB at 45°, and 77.5 dB at 90°. The most effective modification was the combined approach using fitting, blade, and ring optimizations, which reduced axial noise to 78.0 dB and achieved an average sound pressure level of 77.5 dB, representing a 1.7% reduction (1.3 dB decrease) compared to the baseline prototype.

The blade shape optimization configuration also demonstrated improved acoustic performance, with an average noise level of 77.7 dB (1.4% reduction from baseline).

The standard commercial fan exhibited highly directional noise characteristics, measuring 86.0 dB at the 45° angle—significantly higher than any of the prototype designs—but only 75.5 dB at 90°. The commercial fan's average noise level of 80.5 dB was higher than all tested configurations.

All optimized designs showed more uniform noise distribution across measurement angles than the standard commercial fan. The prototypes demonstrated relatively consistent sound pressure levels with variations of less than 3.5 dB between measurement positions, whereas the standard fan showed a variation of 10.5 dB, with a pronounced peak at the 45° position. Most significantly, the optimized prototypes demonstrated a 7.5

dB reduction at the critical 45° measurement angle (from 86.0 dB to 78.5 dB), representing the most substantial improvement observed in this study.

The optimized fitting and closed center designs produced identical noise measurements across all three angles (79.5, 79.5, 76.5 dB).

Thrust measurements showed that the optimized prototype generated the same thrust of 40 grams as the commercial fan.

CONCLUSION

Hypothesis Evaluation: The hypothesis stated: "By optimizing the shape and geometry of fan blades, it is possible to significantly reduce noise pollution without sacrificing airflow efficiency." The experimental results strongly confirm this hypothesis. The fan operated more quietly with no loss of thrust when its blade geometry was optimized with modifications. Specifically, the best-performing design achieved a 1.7% noise reduction (from 78.8 dB to 77.5 dB average) compared to the baseline prototype, and a 3.7% reduction (from 80.5 dB to 77.5 dB average) compared to the standard commercial fan. Most importantly, at the 45° measurement angle—where users typically interact with fans in desk and ceiling configurations—the optimized design achieved a 7.5 dB reduction compared to the commercial fan (from 86.0 dB to 78.5 dB), representing approximately 2.3 times reduction in perceived loudness. These results indicate that aerodynamic shape optimization is an effective approach for reducing fan noise, particularly directional noise peaks, without compromising performance.

Theoretical Explanation Based on Research: These noise reduction results align with established research on aerodynamic blade design. The improvements observed can be attributed to several aerodynamic mechanisms documented in the literature. Research by Chen et al. (2021) demonstrated that leading-edge modifications can facilitate the break-up of vortices passively and effectively without negatively impacting aerodynamic performance. The modified blade geometries in this study likely operated through similar

principles. The geometric optimizations presumably reduced vortex shedding at the trailing edges and minimized turbulent airflow, thereby decreasing tonal noise generation. The more uniform noise distribution across measurement angles in the optimized designs, compared to the highly directional noise pattern of the standard fan, suggests improved aerodynamic characteristics and reduced flow separation.

Analysis of Results: Several results merit detailed discussion. First, the optimized fitting and closed center designs produced identical noise measurements across all three angles (79.5, 79.5, 76.5 dB), suggesting that these modifications may have similar effects on airflow patterns despite different geometric approaches. This convergence warrants further investigation to understand whether these designs produce acoustically equivalent flow fields.

Most significantly, the standard commercial fan exhibited an unusually high noise level of 86.0 dB at the 45° measurement position—6.5 dB higher than the axial measurement and 10.5 dB higher than the 90° measurement. This pronounced directional characteristic suggests concentrated vortex shedding or blade-passage frequency reinforcement at this specific angle, potentially due to blade design, rotational speed, or mounting geometry. The optimized prototypes completely eliminated this directional noise peak. At the 45° angle—where humans typically interact with desk or ceiling fans—the best-performing optimized design achieved a 7.5 dB reduction (86.0 dB to 78.5 dB). Since a 10 dB change represents approximately a doubling or halving of perceived loudness, the 7.5 dB

reduction corresponds to approximately 2.3 times reduction in perceived loudness at this critical angle. This directional noise pattern was not observed in the prototypes, indicating that the geometric modifications specifically addressed the vortex shedding mechanisms responsible for directional noise concentration.

Sources of Error: Several potential sources of error may have affected the accuracy and precision of the measurements:

Shaft and axis alignment: Misalignment issues identified during Phase III testing may have introduced vibration-induced noise of approximately ±1-2 dB, artificially inflating noise measurements across all configurations.

Near-field measurements: Testing at 4 cm distance places measurements in the acoustic near field rather than the far field. Near-field measurements may not accurately represent the far-field acoustic performance that would be experienced in practical applications. Far-field measurements at distances greater than 1 meter would provide more representative data.

Limited sample size: Only three measurement trials per configuration were conducted. Statistical significance would be improved with 10+ trials per geometry to account for measurement variability and establish confidence intervals.

Single RPM testing: All tests were conducted at a single motor speed. Noise characteristics may vary significantly across different RPM ranges, and the optimal geometry for noise reduction at one speed may differ from the optimal geometry at another speed.

Manufacturing variations: Variations in 3D printing quality, including layer adhesion, surface finish, and dimensional accuracy, may have introduced slight geometric differences between nominally identical designs.

Measurement instrument limitations: The smartphone-based sound level meter, while convenient, may have lower accuracy compared to professional-grade sound level meters. Calibration of the measurement device was not verified before testing.

Ambient conditions: Although background noise was measured at 40 dB, variations in temperature, humidity, and air currents throughout the testing period may have affected measurements.

Future Improvements and Extensions: To build upon this research and address identified limitations, several improvements and extensions are recommended:

Anechoic chamber testing: Conduct measurements in an acoustically isolated anechoic chamber to eliminate ambient noise and obtain far-field acoustic measurements at standardized distances (e.g., 1 meter).

Multi-RPM characterization: Test each geometry across a range of rotational speeds from 2,000 to 8,000 RPM to characterize noise-speed relationships and identify optimal operating ranges for each design.

Quantitative thrust measurements: Implement a load cell or force transducer to precisely quantify thrust generation for each configuration, enabling calculation of acoustic efficiency metrics (thrust per unit of noise generated).

Expanded sample size: Conduct 10-15 trials per configuration to enable statistical analysis, including calculation of standard deviations, confidence intervals, and significance testing.

Parametric optimization: Systematically vary geometric parameters (edge treatment depth, spacing, blade twist angle, etc.) to identify optimal configurations and establish design guidelines.

Computational fluid dynamics: Perform CFD simulations to visualize flow patterns, identify vortex formation and shedding, and predict acoustic performance before physical prototyping.

Frequency spectrum analysis: Conduct detailed frequency analysis to identify dominant noise frequencies and assess whether geometric modifications shift noise to higher or lower frequency ranges.

Real-world application testing: Evaluate optimized blade designs in actual applications (computer cooling fans, HVAC systems, drones) to assess practical performance under operational conditions.

Experimental Development Phases: This research progressed through four distinct developmental phases:

Phase I: Initial feasibility testing revealed significant limitations in the test rig. The setup was unable to accelerate the fan to sufficiently high rotational speeds to allow reliable measurements of thrust or sound, necessitating redesign.

Phase II: A revised test setup with new motors and modified structure enabled operation across a wider range of rotational speeds. However, fundamental issues persisted, leading to the decision to completely rebuild the apparatus.

Phase III: The new test rig incorporated a more powerful motor capable of reaching several thousand RPM, enabling the first meaningful measurements. However, shaft and axis alignment problems negatively affected result quality, requiring further mounting system improvements.

Phase IV: After refining the mounting system, comprehensive tests were successfully conducted on multiple modified fan geometries, yielding the data presented in this report.

This iterative development process highlights the importance of experimental apparatus design in obtaining reliable acoustic measurements and demonstrates the challenges inherent in precision noise measurement of rotating machinery.

Conclusion Summary: This research successfully demonstrated that geometric modifications to fan blade design can significantly reduce noise pollution while maintaining aerodynamic performance. The combined optimization achieved the best acoustic performance with a 1.7% average noise reduction compared to baseline and superior performance relative to standard commercial fans. Most significantly, the optimized design achieved a 7.5 dB reduction at the 45° measurement angle compared to the commercial fan, representing approximately 2.3 times reduction in perceived loudness at the angle where users typically interact with desk and ceiling fans. These results validate the application of aerodynamic design principles to engineering

challenges and suggest promising directions for developing quieter fans for applications in computing, HVAC systems, electric vehicles, and unmanned aerial vehicles. The elimination of directional noise peaks represents a major advancement for user comfort in real-world applications. Future work should focus on optimizing geometric parameters, conducting far-field acoustic measurements, and validating performance in real-world applications.

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